Advanced Data Analytics

Innomatics Research Labs — Course Syllabus

NASSCOM FutureSkills Prime Certified Advanced Data Analytics Program. The official page describes a 200+ hour hands-on program organized into five modules.

Source: https://www.innomatics.in/advanced-data-analytics-python-mysql-power-bi-nasscom/

Module 1: Python Programming for Analytics

- Python fundamentals: data types, operators and control flow
- Functions and OOP concepts
- File handling and exceptions
- Reusable and efficient code for data manipulation

Module 2: Exploratory Data Analysis (EDA) using Python

- Descriptive statistics and probability distributions
- NumPy and Pandas data manipulation
- Matplotlib and Seaborn visualization
- Real-world EDA case studies and projects
- Web scraping and data collection exercises

Module 3: Advanced Statistics for Analytics

- Discrete and continuous distributions
- Normal distribution and data transformations
- Sampling, Central Limit Theorem and confidence intervals
- Hypothesis testing using SciPy
- End-to-end analytical applications

Module 4: MySQL for Data Analytics

- SQL syntax, joins and subqueries
- Aggregations and data summarization
- Data extraction and integration with Python
- Analytical use cases using real datasets

Module 5: Power BI and Excel for Data Visualization

- Power BI data cleaning and modeling
- Visuals, KPIs and reports
- DAX functions and measures
- Excel analytics and reporting
- End-to-end Sales Analytics Dashboard project

Prepared from the syllabus/curriculum published on the official Innomatics Research Labs website. Website content can change; use the source URL above as the authoritative current reference.